Data & AI Case Study

Mapped to two Wuerth internship roles

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This is my independent case study. I mapped the requirements of two Wuerth Data & AI internship postings to work I have already built and measured in my own portfolio (now 19+ repositories, including two warehouse-logistics flagships, a supply-network optimizer, an energy forecasting + peak-shaving study, a visual quality-inspection study, and a live installable quantum-explainer PWA). The euro and accuracy figures throughout are on synthetic data (except retail-analytics-real, measured and labelled on real public data) and exist to prove the methods work and are measurable -- not to forecast Wuerth outcomes.

Job #1 -- (Agentic) Automation with Low-code Platforms

Agentic AI workflows, low-code (n8n / Power Automate), connecting systems & APIs, document automation, ROI, rapid prototyping.

Job #2 -- Data & AI Analytics

BI / Power BI, KPI dashboards, forecasting & predictive analytics, Python & SQL, data modelling, turning data into decisions.

Who Wuerth is (public info)

Wuerth (public profile, approximate): a large family-owned German-HQ group in assembly/fastening and industrial MRO distribution -- ~400+ companies in 80+ countries, ~87,000 employees, ~EUR 20B+ revenue, a catalogue in the millions of articles, multi-channel (field sales, branches, e-commerce, e-procurement/EDI, ORSY inventory systems). A business that runs on assortment, pricing, availability, logistics, and high-volume transactional documents -- a natural fit for applied Data & AI.

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Opportunity map (summary)

Wuerth business area (public-info) -> Data/AI approach -> repo + measured result (synthetic unless noted)

Area (public-info)	Approach	Repo	Result (synthetic -- labelled if real)
Procurement / assortment / cross-sell	MILP vs greedy; Apriori/FP-growth	revops-optimizer, market-basket-analysis	MILP > greedy; 254 rules, top lift 2.41
Pricing & margin	Elasticity + leakage; endogeneity fix	sales-kpi-analytics, ml-models-lab	EUR2.6M leakage lever; bias +1.52 -> +0.03
Inventory / replenishment	Newsvendor + forecast	distributor-intel-platform, ml-models-lab	MASE 0.38; MASE 0.987 / RMSSE 0.948 (beats naive + Holt-Winters)
Sales KPIs & analytics	KPIs + rolling-origin CV	sales-kpi-analytics	MASE < 1; exec PDF
Logistics / routing	OR-Tools CP-SAT VRP	route-optimizer	4.6% / 31% savings
E-procurement automation	Agentic + n8n low-code	agentic-automation-lab	~EUR625k/yr modelled
Document processing	Agentic RFQ/invoice extract	doc-extract-agent	~EUR145k/yr modelled
Customer retention	Decline + churn classifiers	revops-optimizer, ml-models-lab	ROC-AUC 0.99; churn ECE 0.197 -> 0.021
Warehouse / intralogistics (new)	Digital twin + slotting + DES + packing	logistics-flow-studio, logistics-digital-twin	-48.6% pick travel; slotting -44.2%; fill 2.0% -> 30.2%
Supply-network design (new)	Facility MILP + flows + safety stock	supply-network-opt	-21.2% cost vs greedy; stock -65.7% / -80.1%
Energy management / facilities (new)	Load forecast (rolling CV) + peak-shaving LP	energy-demand-forecast	MASE 0.497, 14/14 folds (Holt-Winters loses to naive, reported); peak -20.9%; ~EUR11,100/yr at ASSUMED tariff; battery does not pay back on these assumptions
Quality inspection (new)	Clean-only anomaly detection; pre-registered rule	quality-anomaly-vision	AE 0.779 vs PCA 0.772 ROC-AUC -- inside 0.02 margin, PCA recommended; TPR 0.407 vs 0.393 @ 5% FPR
Real data (completed)	Cleaning + RFM + leakage-safe CV	retail-analytics-real	real data: seasonal-naive wins CV, MASE 1.094; Champions 25% -> 69.0%

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Job #1 -- (Agentic) Automation with Low-code

Every posting requirement -> repo + artifact + measured proof (all synthetic data).

Requirement	Repo / artifact	Proof (synthetic unless noted)
Build AI-agent workflows	agentic-automation-lab	agentic tool-use loops
Low-code (n8n / Power Automate)	agentic-automation-lab: n8n/rfq_intake_agent.json	RFQ-intake agent; ~EUR625k/yr
Visual / flow-based building	agent-flow-studio	flow builder; ~EUR47k/yr
Connecting systems / APIs	agentic-automation-lab connector layer	agents call external APIs
Process automation (back-office)	doc-extract-agent	RFQ/invoice -> structured
Document intake / understanding	doc-extract-agent	extract + validate; ~EUR145k/yr
Rapid prototyping	agent-flow-studio, agentic-automation-lab, logistics-flow-studio	runnable prototypes incl. an installable offline PWA
Prototyping + education (hype-free)	quantum-explainer -- LIVE: dimitres-kisimov.github.io/quantum-explainer	hand-written simulator ~300 lines, zero deps; 42 physics assertions + 57 structural checks pass
Show automation ROI	automation-roi-explorer	EUR383k/yr net modelled
Python engineering	all automation repos	Python throughout
Stakeholder communication	automation-roi-explorer	business-readable ROI views

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Job #2 -- Data & AI Analytics

Every posting requirement -> repo + artifact + measured proof (synthetic data unless labelled real data).

Requirement	Repo / artifact	Proof (synthetic unless noted)
BI / Power BI	revops-optimizer: powerbi/DAX_measures.md	DAX pack + 3-page report
KPI dashboards / metrics	sales-kpi-analytics	KPI layer + exec PDF
Predictive analytics / forecasting	sales-kpi-analytics; ml-models-lab global forecaster	MASE < 1; MASE 0.987 / RMSSE 0.948 beats naive + Holt-Winters
Forecasting rigour	distributor-intel-platform	MASE 0.38 (9 rolling folds)
Energy forecasting + optimization	energy-demand-forecast (forecasters + LP from scratch; 19 tests)	MASE 0.497 / MAPE 4.8%, 14/14 folds (Holt-Winters loses to naive, reported); peak 368.2 -> 291.1 kW (-20.9%); ~EUR11,100/yr at ASSUMED EUR12/kW-month; timer baseline EUR0; battery does not pay back on these assumptions
Python for analytics	sales-kpi-analytics, revops-optimizer	end-to-end pipelines
SQL / data modelling	sales-kpi-analytics SQL set	spend analysis queries
Excel-level tabular analysis	sales-kpi-analytics	spend breakdowns
Optimization / prescriptive	revops-optimizer; supply-network-opt; logistics-digital-twin	~EUR160k/yr; -21.2% cost vs greedy; fill 2.0% -> 30.2% (CP-SAT proof)
Elasticity / statistical modelling	revops-optimizer; ml-models-lab	ROC-AUC 0.99; elasticity bias +1.52 -> +0.03; churn ECE 0.197 -> 0.021
Data into business decisions	sales-kpi-analytics; market-basket-analysis	EUR2.6M leakage lever; cross-sell recs, top lift 2.41
Market-basket / cross-sell	market-basket-analysis	Apriori + FP-growth from scratch; 224 itemsets, 254 rules; 18 tests
Classification / anomaly detection	ml-models-lab	SKU macro-F1 0.963; AE PR-AUC 0.963 vs PCA 0.951 (PCA default)
Visual quality inspection (vision)	quality-anomaly-vision (3 detectors, pre-registered rule; 15 tests)	AE ROC-AUC 0.779 vs PCA 0.772 -- inside 0.02 margin, PCA recommended; TPR 0.407 vs 0.393 @ 5% FPR; texture-breaks hard for all (best 0.609)
Warehouse / intralogistics	logistics-flow-studio (WarehouseTwin); logistics-digital-twin	-48.6% pick travel; ABC ~21% > random; slotting -44.2%; DES -76.1% / -66.5%
Logistics / ops analytics	route-optimizer; supply-network-opt	4.6% / 31% savings; safety stock -65.7% / -80.1%
Real, messy data (completed)	retail-analytics-real (UCI Online Retail II, 1,067,371 real rows, CC BY 4.0)	real data: 94.0% rows kept; GBP 19.6M revenue; Champions 25% -> 69.0%; seasonal-naive wins CV, MASE 1.094
Present to decision-makers	revops-optimizer exec deck	exec narrative deck